MIL-C-12475D
19 September 1968
SUPERSEDING
MIL-C-12475C
12 November 1965

MILITARY SPECIFICATION

CHAIR, STRAIGHT, WITHOUT ARMS, WOOD

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size, straight, wooden chair without arms (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-1183	- Plastic Molding Material, Acrylonitrile-Butadiene-Styrene (ABS), Rigid
TT-F-336	- Filler, Wood Paste
TT-L-57	- Lacquer, Rubbing, Clear (For Wood Furniture)
TT-S-190	- Sealer, Sanding, Lacquer Type (For Wood Furniture)
TT-S-300	- Shellac, Cut
TT-S-720	- Stain, Wood, Non-Grain-Raising, Solvent Type
MMM-A-100	- Adhesive, Animal Glue
MMM-A-188	- Adhesive; Urea-Resin-Type (Liquid and Powder)
MMM-A-193	- Adhesive, Vinyl Acetate Resin Emulsion
PPP-B-636	- Box, Fiberboard
* PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

FSC 7105

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MILITARY

MIL-V-12954 - Varnish, Low Bake, Clear Rubbing, For Wood Furniture

STANDARDS

FEDERAL

Fed. Test Method Std. No. 141 - Paint Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing
Fed. Test Method Std. No. 175 - Adhesives; Methods of Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

ARMY NATICK LABORATORIES

5-8-103 - Chair, Straight, W/O Arms
5-8-182 - Chair, Straight, W/O Arms, Alternative Construction

(Miniature copies of Drawing 5-8-103 and 5-8-182, identified as figures 1 and 2 respectively, are attached for information purposes only.)

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurements functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES DEPARTMENT OF COMMERCE COMMERCIAL STANDARDS

Commercial Standard CS 35-61 - Hardwood Plywood

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D 2016-65 - Standard Methods of Test For Moisture Content of Wood.

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(Application for copies of the American Society for Testing Materials Standard Specifications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a preproduction sample, consisting of a sample chair covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

* 3.2 Materials.- Materials not definitely specified shall conform to the quality currently used for the purpose by the manufacturer in commercial practice.

* 3.2.1 Wood.-

* 3.2.1.1 Solid.- The wood shall be uniformly kiln-dried without case-hardening or honeycombing. The wood shall be plain or quarter sawn. The wood components shall be free from defects except that the following will be permitted:

(a) Sound, tight knots not exceeding 1/4 inch in greatest dimension and not less than 10 inches apart.
(b) Sap stain or discoloration, provided that no more than two such defects appear in any one chair.
(c) Not more than two wormholes, 1/16 inch or less in diameter, provided that they are in a concealed surface.

The slope or divergence of grain shall not exceed 1 inch in 10 inches when tested as specified in 4.4.1. The corner blocks shall be sound and free from brashness and decay.

* 3.2.1.1.1 Steam bent wood components.- The steam bent wood components shall be fabricated from any one of the following species of wood:

Rock Elm
Oak
Hard Maple
American Elm
Ceder Elm
Maple, Big Leaf
Hickory
Beech
Birch, Sweet or Yellow
Pecan
Winged Elm

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* 3.2.1.1.2 Wood components other than steam bent components.- All wood components of any one chair, other than steam bent components and band sawn posts, shall be fabricated from one of the woods specified in 3.2.1.1.1 or one of the following species of wood:

Ash, White	Hackberry
Sycamore	Holly
Magnolia	Sweet Gum, Red or Sap

* 3.2.1.2 Laminated plywood.- The plywood shall be of a laminated construction with the grain direction of all plies parallel. The veneer inner plies shall be of either medium or high density, grade 1 or 2, and the face plies shall be grade 1 of Commercial Standard CS 35-61. The veneers shall be full length, shall be no thicker than 1/8 inch and shall be bonded with an adhesive conforming to 3.2.3.2.

* 3.2.2 Plastic molding material.- The plastic molding material shall conform to type II of L-P-1183 and shall be 0.110 inch thick with a plus tolerance 0.005 inch. Clean, unburned plastic molding material in the form of imperfect parts, sprues, runners, or other scraps of the same composition as the virgin material, and produced in the molding or finishing operation, may be reground and mixed with the material specified.

* 3.2.3 Adhesives.-

* 3.2.3.1 Animal glue.- Animal glue adhesive shall conform to either type I, grade J-2, or type II of MMM-A-100.

* 3.2.3.2 Urea-resin-type.- The urea-resin-type adhesive shall conform to either type I or III of MMM-A-188, or the adhesive may be of a thermosetting vinyl emulsion type which meets the block shear strength, tension loading shear strength, both wet and dry, and pH requirements of MMM-A-188.

* 3.2.3.3 Vinyl acetate resin-emulsion.- Vinyl acetate resin emulsion adhesive shall conform to MMM-A-193.

* 3.2.3.4 Aliphatic resin emulsion.- The adhesive shall be aliphatic resin emulsion type, not primarily polyvinyl acetate. This adhesive shall be for use in furniture manufacturing. A representative sample of the adhesive shall meet the following requirements when tested as specified in 4.4.2, 4.4.3, 4.4.4, and 4.4.5.

Properties	Requirements
Block shear strength at room temperature.	2800 psi minimum; 50 percent wood failure minimum average.
Heat resistance (block shear strength at 150 F.).	Block shear strength and wood failure shall be not less than 80 percent of the values obtained at room temperature.
Viscosity at 77 F.	1000 to 7000 centipoise.
Acidity of dried film	pH 5.0 plus or minus 1.5.

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* 3.2.4 Stain.- The stain shall conform to TT-S-720 or stain may be pigmented stain, provided matching of color and uniformity are maintained and provided other coating applications and requirements are met. The pigmented stain shall not prevent the remainder of the lacquer system from meeting the cold check and the toughness and adhesion requirements specified in TT-L-57. The tests of TT-L-57 shall be conducted on panels wherein the pigmented stain has replaced the solvent type stain in the lacquer system.

* 3.2.5 Washcoat.- At the option of the supplier, alternative "A", "B", or "C" washcoat may be used.

* 3.2.5.1 Alternative "A".- Alternative "A" shall consist of shellac varnish conforming to type I, grade B, body 1 of TT-S-300, thinned with 4 parts of denatured alcohol.

* 3.2.5.2 Alternative "B".- Alternative "B" shall consist of lacquer conforming to TT-L-57, reduced for application at 5 to 8 percent solids.

* 3.2.5.3 Alternative "C".- Alternative "C" shall be a laqquer especially formulated for washcoat purpose, to be applied at 5 to 8 percent solids, and to contain no zinc stearate or other sanding aid.

* 3.2.6 Filler, wood paste.- Wood paste filler shall conform to TT-F-336.

* 3.2.7 Lacquer, spray, clear.- The lacquer shall conform to type I or II of TT-L-57, with the exception of the cold check and ultraviolet light resistance test panels. Test panels shall be of the same species of wood as the chair and the other components of the finish shall be as specified in 3.4.1.1 through 3.4.1.6.

* 3.2.8 Sealer.- The sealer shall conform to TT-S-190 except that the panel required for determination of sanding properties shall be of the same species of wood required for the chair and shall be finished as specified in 3.4.1.1 through 3.4.1.5.

* 3.2.9 Varnish, low bake.- The varnish shall conform to MIL-V-12954.

* 3.2.10 Dowels.- Dowels shall be fluted or spirally grooved hardwood. The diameter of the dowels shall conform to Drawing 5-8-103 and Drawing 5-8-182.

* 3.2.11 Glides.- Glides shall be fabricated of corrosion-resisting steel, or case hardened steel with a nickel or chromium protective coating, of the size and construction shown on the applicable drawing. Glides shall be provided with a washer of either natural rubber or synthetic material.

* 3.2.12 Hardware, steel.- The screws, washers, bolts, "T" nails, pins and the corner bracket, when applicable, shall be the current commercial items, shall be zinc or cadmium plated, and shall be of the types and sizes shown on the applicable drawing.

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* 3.3 Construction.- The chairs shall be constructed in accordance with the design, details and dimensions as shown on Drawing 5-8-103 and Drawing 5-8-182. All screws holes shall be predrilled, countersunk or counterbored. As applicable, all exposed surfaces shall be carefully machined and sanded to a smooth finish prior to assembly. The chairs shall not rock more than 1/16 inch when placed in position for normal use on a level surface. Glides shall not become detached when subjected to the pull test specified in 4.4.7. Mortise and tenon joints may be used in lieu of dowels specified in 3.2.10, provided the tenons are at least three fourths of the width of the piece and at least 3/4 inch deep.

* 3.3.1 Solid wood components.-

* 3.3.1.1 Moisture content.- The component parts shall have a moisture content of from 5 to 9 percent at time of assembly, when determined in accordance with 4.3.1.2.

* 3.3.1.2 Seat.- The seat shall be a saddle shaped type and shall be constructed of edge glued, built-up stock with straight joints, tongue and groove joints, or linderman joints, using continuous pieces of wood not less than 1-3/4 inches and not more than 4-1/2 inches wide. Adhesive specified in 3.2.3.2 shall be used on edge gluing. Alternatively, the seat may be constructed in accordance with 3.3.3.1 and as shown on Dwg No. 5-8-182.

* 3.3.1.3 Front legs.- The front legs shall be constructed from one piece of wood and tapered two ways. Outside edges shall be rounded to a minimum 1/4 inch radius. The legs for the alternative construction shall allow for the fit into the recessed area on the underside of the seat.

* 3.3.1.4 Front rail.- The front rail shall be constructed from one piece of wood and shall be double dowelled or mortised and tenoned into the front legs. Alternatively, the front rail may be constructed in accordance with 3.3.3.1 and as shown on Dwg No. 5-8-182.

* 3.3.1.5 Front and back stretcher.- The front and back stretcher shall be constructed from one piece of wood and shall be mortised and tenoned, check tenoned, or dowelled into the legs or back posts.

* 3.3.1.6 Corner blocks.- The chair shall be reinforced with two corner blocks attached to the front and side rails. Each block shall be securely glued and reinforced with two screws, one into each adjoining rail. A screw shall be placed through the corner blocks into each of the front legs. Alternatively, the corner blocks may be constructed in accordance with 3.3.3.1 and as shown on Dwg No. 5-8-182.

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* 3.3.2 Components, steam bent wood or laminated plywood.- The seat rail for the chair (except alternative construction), back posts, back slats and curve stretchers shall be steam bent from one continuous piece of wood of the wood specified in 3.2.1.1.1, or shall be of laminated plywood specified in 3.2.1.2.

* 3.3.2.1 Seat rail.- Seat rail shall be double dowelled into the front legs and secured to the back posts by a lag screw into each post. Alternatively, the seat rail may be constructed in accordance with 3.3.3.1 and as shown on Dwg No. 5-8-182.

* 3.3.2.2 Back slats.- The upper back slat shall be double dowelled into the back posts, or mortised and tenoned. The two lower slats shall be single dowelled, chuck tenoned, or mortised and tenoned into the back posts. All sharp edges of slats shall be removed and the top, inside edges shall be well rounded.

* 3.3.2.3 Curve stretchers.- Curve stretchers shall be mortised and tenoned, chuck tenoned, or dowelled into the front legs and screwed to the back posts at the notched sections.

* 3.3.3 Alternative construction.- The chair shall be furnished in two unassembled parts. The front part shall consist of a plastic seat assembly, two front legs, straight front stretcher and curved stretcher. The back part shall consist of two back posts, three back slats and a straight back stretcher. The front and back stretchers, back slats, and curve stretcher shall conform to 3.3.1.5, 3.3.2.2, and 3.3.2.3, respectively and as specified in Drawing No. 5-8-182.

* 3.3.3.1 Plastic seat assembly.- The seat assembly shall consist of a seat, seat rail, front rail and corner blocks. The seat assembly shall be injection molded in two shells of the material specified in 3.2.2. The shells, bonded as specified in 3.3.6, shall be filled with polyurethane foam in accordance with the manufacturer's commercial practice. The polyurethane foam shall be homogeneous throughout and firm in composition. The cells shall be uniform in size and shall have no blowholes over 1/2 inch in diameter or large accumulations of unexpanded resin or other occlusions. The polyurethane foam shall have a density of not less than 4 pounds per cubic foot when tested as specified in 4.4.8.

* 3.3.4 Application of adhesives.- The adhesives shall be applies in the manner recommended by the manufacturer. Pressure, pressing time, assembly periods, and curing temperature shall conform to the recommendations of the adhesive manufacturer.

* 3.3.5 Adhesives.- The adhesives specified in 3.2.3.1, 3.2.3.2, 3.2.3.3, or 3.2.3.4 shall be applied as specified in 3.3.4, to all joint surfaces except for fabrication of seat (see 3.3.1.2), laminated components (see 3.2.1.2), and for alternative construction.

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* 3.3.6 Alternative construction.- The plastic shells shall be bonded together as specified in 3.3.4 with an adhesive recommended by the adhesive manufacturer. The adhesive material applied on the dowels or chuck tenons used in joints of plastic to wood or wood to plastic and to the dowelled holes shall meet the block shear strength test and the heat resistance test requirements of 3.2.3.4.

* 3.4 Finish.- The surfaces of the chairs shall be finished by either Method A or Method B finish as specified below, except for the plastic seat assembly surfaces. The plastic seat assembly shall have a wood grain etch on all exposed surfaces. A filler material, similar in color with the wood parts, shall be applied to the etched wood grain. An application of urethane lacquer compatible with the plastic base material shall then be applied to the filler. The color shall be micarta, 96-M-34, plank rock maple of one tone with no shading or highlighting. Heads of bolts shall be coated to match the color of the wood. All surfaces of the parts of the chairs, when in the position of normal use, shall appear homogeneous and shall match the color of the sample panel to be furnished by the contracting officer (see 6.3).

* 3.4.1 Method A (air-dry lacquer).-

* 3.4.1.1 Surface preparation.- Prior to finishing, exposed wood surfaces shall have been thoroughly sanded with 5/0 or finer sandpaper. Glue size may be used before the sanding in order to stiffen the wood fibers. Surfaces shall be thoroughly cleaned immediately before application of stain.

* 3.4.1.2 Stain.- The stain specified in 3.2.4 shall be applied to all exposed wood surfaces and allowed to thoroughly dry for not less than 30 minutes. Light colored parts of the piece shall be stained to a uniform color to harmonize with the surrounding surface by the use of sap stain as necessary.

* 3.4.1.3 Washcoat.- A washcoat as specified in 3.2.5 shall be applied to all exposed wood surfaces. After the washcoat has thoroughly air-dried for not less than 30 minutes, the surfaces shall be lightly rubbed with 5/0 or finer sandpaper and all loose particles and dust removed. Drying of stain and washcoat may be expedited by use of auxiliary drying methods not harmfull to the wood or subsequent finish.

* 3.4.1.4 Wood filler.- A paste wood filler specified in 3.2.6 shall be applied to the exposed surfaces of open grained wood. When the filler has dried to a proper wiping consistency, it shall be worked into pores by any means so that all pores are completely filled. Excess filler shall be cleaned from all surfaces, crevices and corners. Filler shall be thoroughly dry before application of subsequent coatings. Drying may be expedited by use of auxiliary drying methods not harmful to the wood or subsequent finish.

* 3.4.1.5 Sealer coat.- A coat of lacquer sealer specified in 3.2.8 shall be applied to all exposed wood surfaces and edges. After the sealer has been thoroughly dried for not less than 1 hour under local air-drying conditions, all exposed sealed surfaces shall be sanded with 5/0 sandpaper (dry) or finer, and all surfaces shall be wiped clean.

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* 3.4.1.6 Lacquer finish.- After application of the sealer coat, a coat of lacquer shall be applied to all exposed wood surfaces. The lacquer shall be as specified in 3.2.7. The coat of lacquer shall be thoroughly air-dried for not less than 1 hour under local conditions. A second coat of the same lacquer shall then be applied to all surfaces and shall be air-dried for not less than 18 hours under local conditions.

* 3.4.2 Method B finish (baked varnish).- Prior to application of the baked varnish finish, exposed wood surface shall be prepared in accordance with requirements in 3.4.1.1 through 3.4.1.4. The prepared surfaces shall then be varnished in accordance with requirements hereinafter specified.

* 3.4.2.1 Varnish.- A coat of varnish specified in 3.2.9 shall be applied to all exposed surfaces. Force drying shall be preceded by a 20 to 30 minute drying period at room temperature or a gradual rise to the force drying temperature. The baking shall be at a maximum surface temperature of 150 F. for a minimum of 2 hours, or a time-temperature cycle recommended by the manufacturer, which must be fully equal in cure to the above time-temperature cycle and approved by the contracting officer. After the item has cooled to room temperature, the surfaces shall be lightly sanded with 5/0 paper and cleaned. A second coat of the same varnish shall be applied to all exposed surfaces and baked in the same manner.

* 3.4.3 Force drying.- A force dry schedule will be permitted in the finishing steps of either Methods A or B provided performance requirements specified in 3.4.6 are complied with.

* 3.4.4 Rubbing.- Surfaces finished in accordance with methods A and B shall then be rubbed to a uniform satin finish similar in color to the sample panel (see 6.3) and then cleaned. Care shall be exercised not to cut through the finish.

* 3.4.5 Concealed surfaces.- The underside of the seat and the inside and the underside of rails for the chair (except alternative construction) and the underside of the stretchers shall receive not less than one coat of lacquer sealer specified in 3.2.8 for method A finish, or one coat of varnish specified in 3.2.9 for method B finish.

* 3.4.6 Finish resistance.- All surfaces which have been rubbed as specified in 3.4.4 shall show no signs of print or indentation, permanent spotting, or softening when tested as specified in 4.4.6, 4.4.6.1, 4.4.6.2, and 4.4.6.3. Spotting which can be removed with a clean, damp cloth, or softening which hardens within 1 hour shall be acceptable.

* 3.5 Identification marking.- The chair shall be marked on the underside of the seat with the manufacturer's identification or readily identifiable trademark or both.

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* 3.6 Assembly instructions.- One set of comprehensive assembly instructions shall be furnished with each alternative construction chair.

* 3.7 Assembly.- The two parts of the alternative construction chair shall have holes drilled or punched for easy assembly with screws, bolts, nuts, pins and washers, as applicable. The assembly of the components shall be readily accomplished as specified in 4.4.9 without special tools.

* 3.8 Workmanship.- The end items shall be free of rough or sharp edges and splinters. Components shall not be broken, cracked, split or malformed. All exposed surfaces shall be smooth and free of tool marks, sink marks, cuts, dents, blistering cracks, runs, sanding scratches grit, evidence of excessive adhesive and free from imbedded foreign matter. The natural wood grain shall not be clouded by finishing materials.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects as specified in table I, dimensions specified, and tested as specified in 4.4.6, 4.4.6.1, 4.4.6.2, 4.4.6.3, and 4.4.7, and 4.4.9, as applicable.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.3.1.1 Testing of adhesives.- The adhesive requirements of 3.2.3 shall be tested as specified in 4.4.2, 4.4.3, 4.4.4 and 4.4.5. The sample unit shall be one laminated specimen equal in construction to any laminated component. Failure of any one of the above tests shall be cause for rejection of the lot of adhesive offered for testing.

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* 4.3.1.2 Testing for moisture content of wood.- Moisture content of wood requirement in 3.3.1.1 shall be determined in accordance with ASTM-D 2016-65 at time of assembly, except that when testing is performed by the electric moisture meter method, three determinations shall be made on each sample unit. The average of the three determinations shall be the moisture content of the sample unit. The lot shall be all machined components prepared under the same conditons, stored as a group, and offered for inspection at one time The sample unit shall be one component. The inspection level shall be S-1 with an acceptable quality level (AQL) of 4.0, expressed in terms of defects per hundred units.

* 4.3.1.3 Testing for slope of wood grain.- The slope of wood grain requirement specified in 3.2.1.1 shall be determined in accordance with 4.4.1. The lot shall be all the solid wood offered for inspection at one time, prior to assembly. The sample unit shall be one piece of solid wood prior to machining. the inspection level shall be S-1 with an AQL of 4.0, expressed in terms of defects per hundred units.

* 4.3.1.4 Foam density test.- The foam density requirements of 3.3.3 shall be determined as specified in 4.4.8. The sample unit shall be one plastic seat. Failure to comply shall be cause for rejection of the lot of plastic seats offered for testing.

* 4.3.2 Intermediate examination.- Examination shall be made of the chairs for defects in table I. The sample unit for this inspection shall be one chair, prior to the application of the finish. The inspection level shall be II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects (intermediate inspection)

Examine	Defect	Classification Major	Minor
Wood components	Not specified species of wood.	X	
	Open glue joint in edge glued member.	X	
	Not machined or sanded to a smooth finish, as applicable.	X	
	Open or through shake, or honeycomb.	X	
	Loose or unsound knot.	X	
	Sound, tight knot over 1/4 inch in greatest dimensions, or two sound, tight knots not over 1/4 inch less than 10 inches apart.		X
	More than two sap stains or discoloration in one chair.		X

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TABLE I.- Classification of defects (intermediate inspection) (cont'd)

Examine	Defect	Classification Major	Minor
Wood components (cont'd)	Worm hole over 1/16 inch diameter or more than two worm holes 1/16 inch diameter or less in concealed surfaces.		X
	Worm hole on exposed surface.	X	
Solid wood legs, front rails, front and back stretcher	More than one piece of wood.	X	
Legs	Not tapered two ways, as specified.	X	
Construction, assembly, and workmanship	Screw point protrudes through, bulges or splits wood in seat.	X	
	Screw point protrudes through, bulges or splits wood in seat rail.	X	
	Corner block split from screw.		X
	Receiving hole for screw not predrilled, countersunk or counter-bored where required.		X
	Component not attached as required, in wrong location or visibly out of alignment.		X
	Wood component not glued where required.	X	
	Any loose or unsound glued wood joint.	X	
	Joint not specified type.	X	
	Component warped, twisted or malformed from specified shape.		X
	Seat not saddle shaped type.	X	
	Excess glue not removed from assembly joints or from glue joints on exposed surfaces.		X
	Component not steam bent from one continuous piece where required.	X	
	Metal corner bracket not zinc or cadmium plated (if applicable).		X
	Metal corner bracket loose or improperly positioned (if applicable).		X
	Wood corner block loose or improperly positioned (if applicable).		X

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* 4.3.3 Process inspection.- Inspection shall be made during the manufacturing process to determine compliance with 3.3. Whenever nonconformance is noted, correction shall be made to affected items and process.

* 4.3.4 Inspection of the end item.- The lot shall be the number of chairs of the same construction offered for inspection at one time. The sample unit for this inspection shall be one complete chair.

* 4.3.4.1 Visual examination.- Examination shall be made of the chairs for defects in table II. The inspection level shall be III with an (AQL) of 2.5 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE II.- Classification of defects (end item)

Examine	Defect	Classification Major	Minor
Finish and appearance of exposed surfaces	Does not match sample.	X	
	Color not homogeneous, i.e., has two-tone effect, shading or high-lighting visible at 5-feet.	X	
	Area of no film.		X
	Excess filler material not removed from surfaces, crevices or corners.		X
	Peeling, flaking, blistered, cracked or checked.	X	
	Wrinkles, mar, dent, light scratch or orange peel visible at 5 feet.	X	
	Heavy scratch.	X	
	Overall open grain (filler or finish material does not fill wood pores).	X	
	Finish tacky (not completely dry).	X	
	Dirt, grit, or foreign matter imbedded in finish.		X
	Unstained area (white wood visible or light colored edges not touched up) visible at 5 feet.		X
	Component not stained where required or general flooding of stain.	X	
	Not lacquer or varnish coated (as applicable).	X	
Glide and attachment of glide	Broken, cracked, fractured, or malformed.		X
	Rubber insert cut, split, damaged, or missing.		X

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TABLE II.- Classification of defects (end item) (cont'd)

Examine	Defect	Classification Major	Minor
Glide and attachment of glide (cont'd)	Plating on glide peeling, loose, flaking, or cut through to metal.		X
	Not type specified.		X
	Improperly positioned or does not fit properly (loose, cocked, or crooked on leg).		X
	Missing.	X	
Screws, lag screws, bolts, and washers	Not specified type.		X
	Lag bolt missing.	X	
	Carriage bolt missing.	X	
	Washer missing.		X
	Screw missing (except in bent U-stretcher).		X
	Screw missing in bent U-stretcher.	X	
	Screw thread stripped in wood, or more than one screw not drawn up tight.		X
Plastic seat	Foam filler plug missing.	X	
	More than one vent hole missing.	X	
Assembly instructions	Missing, illegible, or not comprehensive.	X	
Identification marking	Missing, illegible, incorrect, incomplete, in wrong location.		X

* 4.3.4.2 Assembly instructions.- Examination shall be made of an alternative construction chair for defects in assembly and fit of components for compliance with 3.10 when tested as specified in 4.4.9. The inspection level for this examination shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

* 4.3.4.3 Dimension examination.- Inspection shall be made of the end item for defects in dimensions. Any dimension that is not within the specified tolerance shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

* 4.3.5 Testing of the end item.- Testing of the end item shall be performed in accordance with 4.4.6, 4.4.6.1, 4.4.6.2, 4.4.6.3, 4.4.7 and 4.4.9. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

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* 4.3.6 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be as indicated in table III. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be all of the containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE III.- Examination of preparation for delivery

Examine	Defect
Markings	Omitted, incorrect, illegible, of improper size, location sequence, or method of application.
Materials	Component missing; damaged.
Content	Number per container more or less than required.

* 4.3.6.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure and strapping shall be in accordance with the appendix of that specification.

4.4 Tests.-

4.4.1 Slope of wood grain.- The slope of grain may be determined by combing the grain direction of two adjacent faces. When one of these faces is straight grained, the true slope of grain is shown on the other face. When there is a slope on both faces, the true or combined slope is greater than the slope on either face. Slope of grain is shown on edge grained faces by the summerwood bands, by the direction in which a free flowing ink or dye spreads, or by the course taken by a narrow strip lifted by the knife point or torn out. Direction of fiber may also be measured by the Teco Slope of Grain Detector supplied by the Timber Engineering Company, Inc.

* 4.4.2 Block shear strength test.- Block shear strength test of either wood or plastic components, as applicable, shall be determined in accordance with Method 1031 of Fed. Test Method Std. No. 175.

* 4.4.3 Heat resistance test.- Heat resistance test of either wood or plastic components, as applicable, shall be determined in accordance with Method 1031 of Fed. Test Method Std. No. 175, except conditioned specimens to be heated at 150 F. plus or minus 2 F. for 16 hours plus or minus 1 hour in a forced air circulation oven. Each specimen shall be tested while still hot and immediately upon removal from the oven.

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4.4.4 Viscosity test.- Viscosity at 77 F. shall be determined per Brookfield Viscosity test procedure described in MMM-A-193, except that alternatively, an LVF Brookfield Viscometer and No. 3 spindle at 12 rpm may be used.

4.4.5 Acidity of the dried film test.- The acidity of the dried film shall be determined with Method 4011 of Fed. Test Method Std. No. 175.

* 4.4.6 Finish tests.- Prior to the testing specified in 4.4.6.1, 4.4.6.2, and 4.4.6.3, finishes shall be thoroughly cleaned with xylol or mineral spirits.

* 4.4.6.1 Print resistance test.- The finish of the chair shall be tested in accordance with Method 6211 of Federal Test Method Standard No. 141. The test shall be made not less than 4 hours after rubbing the finish. A pressure of 1 psi shall be applied for not less than 4 hours. The finish shall then be examined to determine compliance with 3.4.6.

* 4.4.6.2 Spot resistance test.- One pool of cubic centimeters of 50-percent pure ethyl alcohol shall be placed on the finish of the chair and covered with a small watch glass over which a larger watch glass shall be superimposed. Tape shall be placed around the contact surface of the large watch glass and finish to form an air tight enclosure. After 18 hours, the alcohol shall be removed by clear water and the finish cleaned with a clean cloth. After 1 hour, the finish shall be examined to determine compliance with 3.4.6.

* 4.4.6.3 Boiling water resistance test.- Not less than 25 cubic centimeters of boiling water shall be poured on the finish seat of the chair. Water shall be allowed to cool to room temperature. The water shall then be removed from the finish by a clean cloth and the finish examined to determine compliance with 3.4.6.

* 4.4.7 Glide securing test.- A glide which has been inserted into proper position in the leg of a chair shall be subjected to a 20-pound pull test, and inspected for compliance with 3.3.

* 4.4.8 Foam density test.- The bonded shells shall be weighed prior to and after filling with foam. The net weight of the foam shall be as specified in 3.3.3.

* 4.4.9 Assembly test.- Components specified in 3.3.3 of the alternative construction chair shall be assembled. The assembling shall comply with 3.10.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

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* 5.1.1.1 Disassembly.- Each alternative construction chair shall be disassembled into two parts: one, the back section assembled, comprised of the two back posts, the three back slats and the straight back stretcher; two, the front section assembled, comprised of the seat assembly, the two front legs, straight front stretcher and the curved stretcher.

* 5.1.1.2 Cushioning.- Each chair shall have all the surfaces which contact the inner surfaces of the shipping container or each other covered with cushioning material. Cushioning material shall be fabricated in the form of pads or blankets and may be any one of the following materials or combination thereof to provide built-up cushioning of not less than 3/8-inch thickness:

Macerated paper, plain	Cellulose wadding
Macerated paper, cellulose faced	Excelsior (sealed ends)
Shredded paper	Cotton wadding
Wood wool	Molded pulpboard
Wood fiber felt	Laminated cellulose wadding backed with corrugated fiberboard

All cushioning shall be secured to the chair by spiral ties of jute, sisal, manila, or cotton twine having a minimum breaking strength of 30 pounds, the spiral windings being spaced not more than 2 inches apart. The twine shall be securely tied to maintain cushioning in position.

* 5.1.1.3 Assembled chairs.- Two assembled chairs shall be packaged together with one chair inverted so that the seat of one rests on the seat of the other. The front legs of each chair and the adjacent back leg posts of the opposing chair shall be tied together tightly with twine and the knot locked securely. The twine shall have a minimum breaking strength of 80 pounds.

* 5.1.1.4 Disassembled chairs.- Hardware required for assembly of one disassembled chair, together with one set of assembly instructions, shall be placed in a cotton cloth drawstring bag and the bag securely tied to the front section of each chair. Four disassembled chairs shall be packaged together. One front section shall be positioned with the legs horizontal and the front rail down. The second front section shall be nested on top of the first section with the legs horizontal, front rail up and the seat at the opposite end. The remaining two front sections shall be nested in the same manner and placed alongside the first nest with the seat of one nest facing the seat of the other nest. The four back sections shall be laid flat on top of the four nested front sections.

5.1.2 Level C.- Chairs shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

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5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.-

* 5.2.1.1 Assembled chairs.- Two assembled chairs, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FOL, grade V2s of PPP-B-636. The two chairs shall be positioned within the container so that the backs abut against the sides of the container.

* 5.2.1.1.1 Alternatively, fiberboard shipping containers may be style RSC, except that when such containers are used, 4-piece wood frames shall be provided at the top and the bottom of each container. The frames shall be not less than 5/8 inch in thickness and of sufficient width to prevent the chair legs from shifting off the frames while in transit. Frames shall be clinch nailed with not less than three nails at each point of attachment. Chair legs shall not be attached to frames by nails or screws.

* 5.2.1.2 Disassembled chairs.- Four disassembled chairs, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FOL, grade V2s of PPP-B-636.

* 5.2.1.3 Closure, waterproofing by means of tape, and reinforcing shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B.-

* 5.2.2.1 Assembled chairs.- Two assembled chairs, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FOL, type CF, variety SW, or type SF, class domestic, grade 275 of PPP-B-636. The body joint of the shipping container shall be lapped and secured with metal fastenings. The two chairs shall be positioned within the container so that the backs abut against the sides of the container.

* 5.2.2.1.1 Alternatively, fiberboard shipping containers may be style RSC, except that when such containers are used, 4-piece wood frames shall be provided at the top and bottom of each container. The frames shall be not less than 5/8 inch in thickness and of sufficient width to prevent the chair legs from shifting off the frames while in transit. Frames shall be clinched nailed with not less than three nails at each point of attachment. Chair legs shall not be attached to frames by nails or screws.

* 5.2.2.2 Disassembled chairs.- Four disassembled chairs, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FOL, type CF, variety SW, or type SF, class domestic, grade 275 of PPP-B-636.

* 5.2.2.3 Closure shall be in accordance with method II of the appendix of PPP-B-636.

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* 5.2.2.4 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto. The shipping container material may also be grade V4s of PPP-P-320.

* 5.2.3 Level C.- Chairs, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use.- The chairs covered by this specification are used in Military dining halls with the dining table covered by MIL-T-12474.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) First article.- When preproduction sample is not required (see 3.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.4).

* 6.3 Sample panels of the required color may be obtained from the procuring activity (see 3.4).

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - MD
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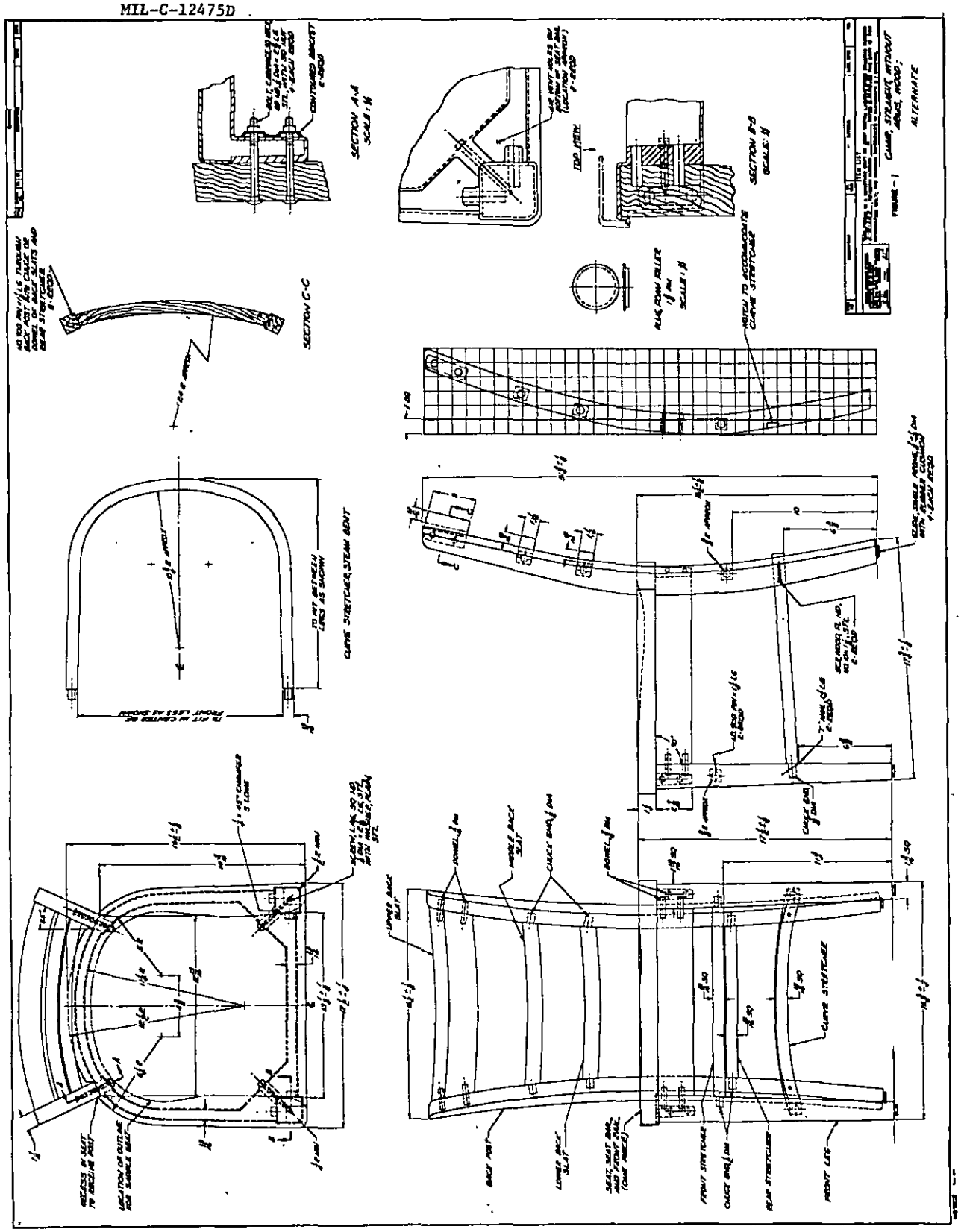
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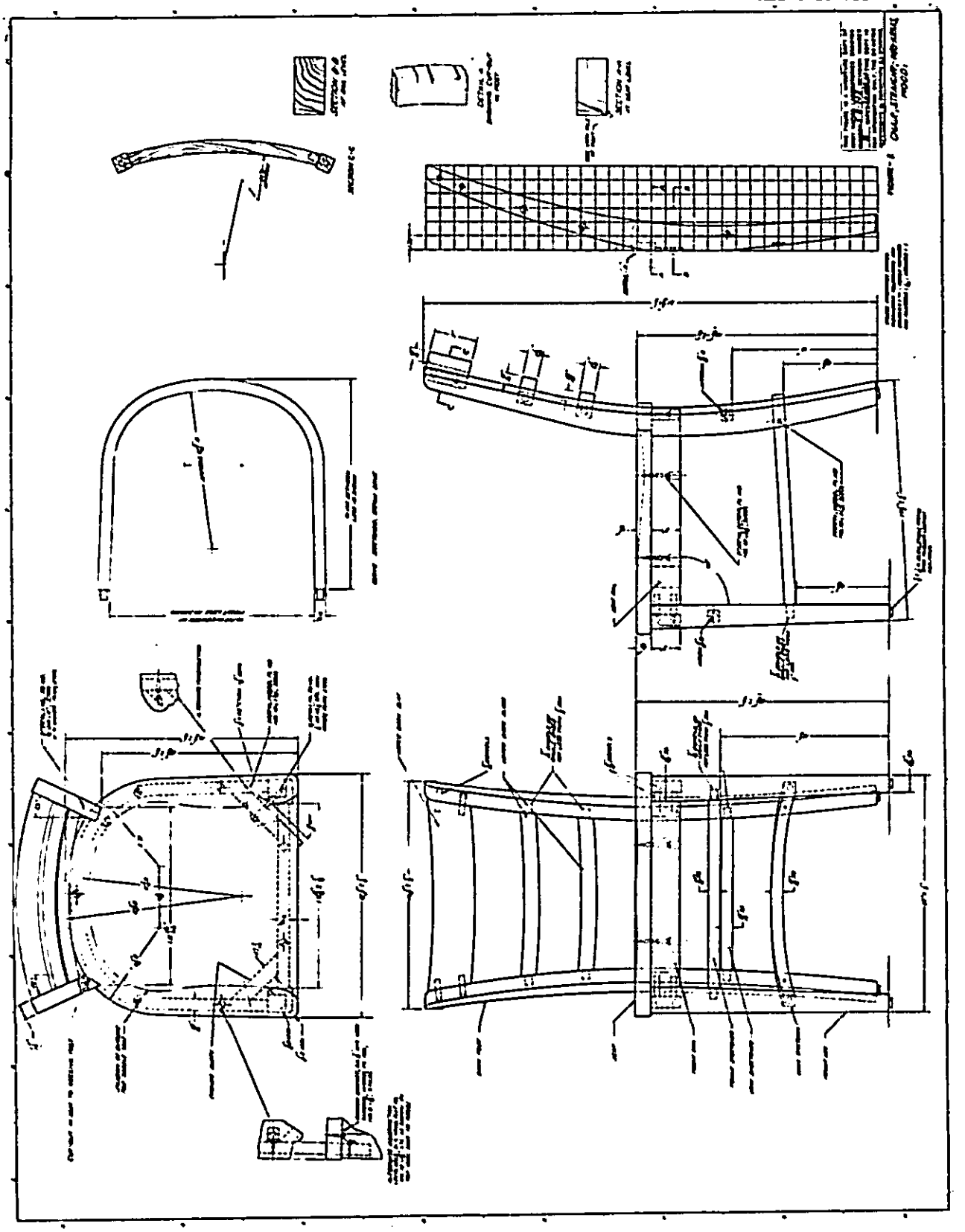
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Chair, Straight, Without Arms, Wood MIL-C-12475D

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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